

Basic Facts/Identities

- $x^2 \geq 0$,
- $2xy \leq x^2 + y^2$ ($x, y \geq 0$),
- Pythagoras' Theorem, Hero's Formula, Ceva's Theorem,
- $|\Delta ABC| = \frac{1}{2}|AB||BC|\sin \theta$, where $\theta = \angle ABC$.

Some Problems

1. (2nd Ir. M. O., Problem) Take a square with side length 1 and choose points A, B, C , and D , one on each side consecutively. Prove that

$$2 \leq |AB|^2 + |BC|^2 + |CD|^2 + |DA|^2 \leq 4.$$

HINT: draw a picture, What shapes do you see?

2. (Putnam, 1998 A6) Let A, B and C denote distinct points with integer coordinates in $\mathbb{R}^2$. Prove that if

$$(|AB| + |BC|)^2 < 8|\Delta ABC| + 1,$$

then ABC are three vertices of a square.

Hint: Prove firstly $4|\Delta ABC| < 2|AB||BC|$.

3. (Putnam 2006 B1) Show that the curve $x^3 + 3xy + y^3 = 1$ contains only one set of three distinct points, A , B , and C , which are vertices of an equilateral triangle, and find its area.

HINT: factor out a linear term $ax + by + c$. Then the curve is this line multiplied by a cubic. What must c be? (guess it is an integer). Then put $a = b = 1$ and do long division.

4. (Toronto Undergraduate Math Comp. 2021) Let a, b, c, d be the lengths of the sides of a quadrilateral. Suppose that the sum of any three side-lengths is an integer multiple of the fourth side. Prove that two of the side-lengths are equal.

5. (Putnam, 2009 A1) Let f be a real-valued function on the plane such that for every square $ABCD$ in the plane, $f(A) + f(B) + f(C) + f(D) = 0$. Does it follow that $f(P) = 0$ for all points P in the plane?

HINT: Draw P in the center of a square. Chop it up into smaller squares, and apply the assumption to all of them.